Algebraic Independence in the Affine Case of Erdős Problem 270

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Erdős Problem 270 asks whether the series of reciprocals of $(n+1)(n+2)\cdots(n+f(n))$ must be irrational whenever $f(n) \to \infty$. Crmarić and Kovač showed that the answer is negative without a regularity assumption, while the problem remains open for nondecreasing f. Erdős and Graham singled out the simplest such choice, $f(n) = n$. We study the entire positive affine subfamily $f(n) = an + b$, whose members are

$$C_{a,b} := \sum_{n=1}^{\infty} \frac{n!}{((a+1)n+b)!}$$

for $a \geq 1$ and $b \geq 1 - a$. For each fixed a, we prove that $C_{a,0}, \ldots, C_{a,a-1}, C_{a,a+1}$ are algebraically independent over $\overline{\mathbb{Q}}$ and that every other admissible $C_{a,b}$ is a nonconstant rational affine combination of them. Thus the fixed-slope family has transcendence degree $a + 1$; in particular, every positive affine case of Problem 270 is transcendental. We begin with an elementary irrationality proof for $0 \leq b \leq a$, which includes the Erdős–Graham example. A Gaussian integral representation then gives a short transcendence proof for that example. For the general theorem, we pass to hypergeometric E-functions and combine algebraic-independence results of Salikhov and Salikhov–Viskina with a logarithmic obstruction at infinity and Beukers's specialization theorem.

1 Irrationality in the affine family

Erdős and Graham ended their chapter on irrationality and transcendence with the question now known as Erdős Problem 270. Given a positive integer-valued function f with $f(n) \to \infty$, they considered

$$S(f) := \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)\cdots(n+f(n))},$$

and asked whether $S(f)$ must be irrational. They expected an affirmative answer when f is nondecreasing [1, p. 66]. Crmarić and Kovač later showed that some regularity assumption is necessary: as f ranges over all such functions, $S(f)$ takes every value in $(0, \infty)$ [3, Theorem 1]. The nondecreasing problem remains open [2].

For integers $a \geq 1$ and $b \geq 0$, the affine choice $f(n) = an + b$ gives

$$S(f) = \sum_{n=1}^{\infty} \frac{1}{(n+1)\cdots((a+1)n+b)} = \sum_{n=1}^{\infty} \frac{n!}{((a+1)n+b)!} =: C_{a,b}.$$

Erdős and Graham singled out the simplest choice, $f(n) = n$, and described the resulting series as difficult to treat [1, pp. 65–66]. It is the case $a = 1$, $b = 0$ of the family above. The following argument proves irrationality throughout the larger range $0 \leq b \leq a$.

Fix $a \geq 1$ and $0 \leq b \leq a$, and put

$$D_n := \frac{((a+1)n+b)!}{n!}.$$

Then

$$\frac{D_{n+1}}{D_n} = \frac{\prod_{j=1}^{a+1}\Big((a+1)n+b+j\Big)}{n+1}.$$

The factor with $j = a+1-b$ is $(a+1)(n+1)$. Cancelling $n+1$ gives the positive integer

$$r_n := \frac{D_{n+1}}{D_n} = (a+1) \prod_{\substack{1 \leq j \leq a+1 \\ j \neq a+1-b}} \Big((a+1)n+b+j\Big).$$

Thus $D_n \mid D_{n+1}$. Moreover, $r_n \geq a+1 \geq 2$, and every factor in the product defining r_n increases with n. Hence r_n is increasing and tends to infinity. Let

$$P_N := \sum_{n=1}^{N} \frac{1}{D_n}, \qquad R_N := C_{a,b} - P_N.$$

Since $D_1 \mid D_2 \mid \cdots \mid D_N$, we have $D_N P_N \in \mathbb{Z}$. Moreover,

$$\begin{aligned} 0 < D_N R_N &= \sum_{k=1}^{\infty} \frac{1}{r_N r_{N+1} \cdots r_{N+k-1}} \\ &\leq \sum_{k=1}^{\infty} \frac{1}{r_N^k} = \frac{1}{r_N - 1} \longrightarrow 0. \end{aligned}$$

If $C_{a,b} = p/q$, then $D_N R_N$ is positive and belongs to $q^{-1}\mathbb{Z}$, so it is at least $1/q$. This is impossible for large N. Hence $C_{a,b}$ is irrational for every $a \geq 1$ and $0 \leq b \leq a$.

Corollary. *The Erdős–Graham series*

$$C_{1,0} = \sum_{n=1}^{\infty} \frac{n!}{(2n)!}$$

is irrational.

Crmarić and Kovač returned to this example and reported that they knew no proof of its irrationality [3, p. 55, following (1.1)]. After their paper appeared, they found the $a = 1$, $b = 0$ specialization of the argument above and suggested that similar reasoning might extend further; Kovač posted this observation on the Erdős Problems forum in July 2026 [4]. We obtained the general argument independently before learning of their post.

2 Transcendence of $C_{1,0}$

Crmarić and Kovač record the Gaussian integral representation [3, p. 55 and footnote 1], obtained by expanding $e^{1/4-t^2}$ and integrating termwise:

$$C_{1,0} = e^{1/4} \int_0^{1/2} e^{-t^2}\, dt. \tag{1}$$

In a 2023 MathOverflow discussion of the neighboring integral $\int_0^1 e^{-t^2}\, dt$, Weber and Pinelis [6] considered the E-function

$$y(x) := \frac{1}{\sqrt{x}} \int_0^{\sqrt{x}} e^{-t^2}\, dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \frac{x^n}{n!}$$

and used the Siegel–Shidlovsky theorem to prove that $y(1)$ and e^{-1} are algebraically independent. The differential system in their argument is ordinary at every nonzero algebraic point, so applying the same theorem at $x = 1/4$ gives the algebraic independence of $y(1/4)$ and $e^{-1/4}$. Since

$$y(1/4) = 2 \int_0^{1/2} e^{-t^2}\, dt, \qquad C_{1,0} = \frac{y(1/4)}{2e^{-1/4}},$$

an algebraic value of $C_{1,0}$ would give a nontrivial algebraic relation between $y(1/4)$ and $e^{-1/4}$. Hence $C_{1,0}$ is transcendental. The MathOverflow discussion did not mention either the factorial series or this specialization.

We now turn to the hypergeometric E-functions associated with the full affine family.

3 Algebraic independence in the affine family

Theorem 1. *For every integer $a \geq 1$, the numbers*

$$C_{a,0}, C_{a,1}, \ldots, C_{a,a-1}, C_{a,a+1}$$

are algebraically independent over $\overline{\mathbb{Q}}$. Every $C_{a,b}$ with $b \geq 1-a$ is a nonconstant rational affine combination of these $a+1$ numbers, and

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(C_{a,b} : b \geq 1-a) = a + 1.$$

Consequently every $C_{a,b}$ in this range is transcendental.[1]

A logarithmic obstruction at infinity gives functional independence; we then specialize at $z = 1$.

[1]An earlier version of this paper established only individual transcendence. Christopher D. Long pointed out that the same machinery should also yield algebraic independence [5]; the argument was then strengthened to the form given here.

For integers $a \geq 1$ and $b \geq 0$, define

$$F_{a,b}(z) := b! \sum_{n=0}^{\infty} \frac{n!}{((a+1)n+b)!} z^{an}. \tag{2}$$

Then

$$F_{a,b}(1) = 1 + b! C_{a,b}. \tag{3}$$

3.1 The first a functions

Fix $a \geq 1$, write $F_b = F_{a,b}$, and let $\theta = z\,d/dz$. Put

$$H := F_{a-1}, \qquad J_a := \{1, \ldots, a+1\} \setminus \{2\}, \qquad \beta_j := \frac{a-1+j}{a+1} \quad (j \in J_a).$$

The multiplication formula gives

$$H(z) = {}_1F_a\left(\begin{matrix} 1 \\ (\beta_j)_{j \in J_a} \end{matrix} ; \frac{z^a}{(a+1)^{a+1}} \right). \tag{4}$$

Indeed,

$$\prod_{j=1}^{a+1} \left(\frac{a-1+j}{a+1} \right)_n = \frac{((a+1)n+a-1)!}{(a-1)!(a+1)^{(a+1)n}},$$

and the omitted factor is $(1)_n = n!$.

To apply the results of Salikhov and Salikhov–Viskina, set

$$\rho_a := \frac{a}{(a+1)^{(a+1)/a}}, \qquad \lambda_j := \beta_j - 1 = \frac{j-2}{a+1},$$

and

$$\varphi(w) := \sum_{n=0}^{\infty} \frac{(w/a)^{an}}{\prod_{j \in J_a} (\lambda_j + 1)_n}.$$

Then $H(z) = \varphi(\rho_a z)$. This hypergeometric E-function is the $l = 0$, $t = a$ case of Salikhov's formula [8, eq. (1)].

Suppose first that a is odd. Salikhov's Theorem 1 applies provided that the parameter multiset modulo integers is not invariant under translation by $1/d$ for any divisor $d > 1$ of a [8, Theorem 1]. Here that multiset is

$$S_a = \left\{ \frac{1}{a+1}, \ldots, \frac{a}{a+1} \right\} \subset \mathbb{Q}/\mathbb{Z}.$$

If $d > 1$ divides a, then $\gcd(d, a+1) = 1$, so $1/d$ does not belong to the subgroup $(a+1)^{-1}\mathbb{Z}/\mathbb{Z}$ containing S_a. Translation by $1/d$ therefore cannot preserve S_a. Salikhov's theorem gives algebraic independence of the values of φ and its first $a-1$ derivatives at every nonzero algebraic point.

When a is even, Salikhov–Viskina show that algebraic dependence of the corresponding function and its first $a-1$ Euler derivatives requires one of the shifts λ_j to be an integer [9, p. 833]. None of the numbers $(j-2)/(a+1)$ with $j \in J_a$ is an integer, so the same functional algebraic independence holds in the even case.

For odd a, the value theorem also implies functional independence. A relation over $\mathbb{C}(z)$ among functions with algebraic Taylor coefficients descends to $\overline{\mathbb{Q}}(z)$: after fixing the monomials and degree bounds in a hypothetical relation, comparison of Taylor coefficients gives a homogeneous linear system over $\overline{\mathbb{Q}}$ in its polynomial coefficients. Specialization at a nonzero algebraic point where those coefficients are defined and not all zero would then contradict Salikhov's value theorem. Since ordinary and Euler derivatives are related by an invertible triangular transformation, we have in every parity

$$H, \theta H, \ldots, \theta^{a-1} H \quad \text{algebraically independent over } \mathbb{C}(z). \tag{5}$$

Direct coefficient comparison in (2) gives the contiguous relation

$$F_{b-1} = F_b + \frac{a+1}{ab}\theta F_b \qquad (1 \le b \le a). \tag{6}$$

The coefficient of z^{an} on the right is

$$\frac{b!n!}{((a+1)n+b)!}\left(1 + \frac{(a+1)n}{b}\right) = \frac{(b-1)!n!}{((a+1)n+b-1)!}.$$

Iterating (6) gives

$$F_j = P_j(\theta)H, \qquad P_j(X) := \prod_{k=j+1}^{a-1}\left(1 + \frac{a+1}{ak}X\right), \qquad 0 \le j \le a-1, \tag{7}$$

where an empty product is 1. The polynomials $P_{a-1}, P_{a-2}, \ldots, P_0$ have degrees $0, 1, \ldots, a-1$ and nonzero leading coefficients. Thus (7) is an invertible triangular transformation, and (5) implies that

$$F_0, F_1, \ldots, F_{a-1} \tag{8}$$

are algebraically independent over $\mathbb{C}(z)$. Let

$$\mathcal{K}_a := \mathbb{C}(z)(H, \theta H, \ldots, \theta^{a-1} H).$$

Then $\mathcal{K}_a$ has transcendence degree a over $\mathbb{C}(z)$. Its stability under differentiation follows from the hypergeometric equation (11) below.

3.2 The boundary function

Let $G := F_{a+1}$. The multiplication formula gives

$$G(z) = {}_2F_{a+1}\left(\begin{matrix} 1, 1 \\ \left(1 + \frac{j}{a+1}\right)_{1 \le j \le a+1} \end{matrix}; \frac{z^a}{(a+1)^{a+1}}\right), \tag{9}$$

All parameters in (9) are positive rational numbers, and with two upper and $a+1$ lower parameters the required pullback degree is a. Thus G is a hypergeometric E-function. Comparing coefficients in (2) gives

$$\left(\frac{\theta}{a}+1\right)G=(a+1)!z^{-a}(F_0-1). \tag{10}$$

We next compare the formal expansions of the two sides at infinity. Put

$$\kappa_a:=(a+1)^{-(a+1)},\qquad P_a(u):=\prod_{j\in J_a}(u+\lambda_j),\qquad c_a:=P_a(0)\neq 0.$$

The hypergeometric equation for H is

$$\left[P_a\left(\frac{\theta}{a}\right)-\kappa_a z^a\right]H=c_a. \tag{11}$$

Lemma 3.1 (Logarithmic boundary obstruction)**.** *The formal image of $\mathcal{K}_a$ at infinity lies in a differential field containing no power of* $\log z$*. The zero-exponential component of every formal solution G of (10), however, contains*

$$-a(a+1)!z^{-a}\log z. \tag{12}$$

In particular, $G\notin\mathcal{K}_a$.

Proof. Set $x=z^{-1}$ and $K=\mathbb{C}((x))$. Adjoin the constant solution to (11) and write the resulting rank-$(a+1)$ companion vector as

$$Y_0=1,\qquad Y_j=\theta^{j-1}H\quad(1\le j\le a).$$

A zero-exponential formal particular solution has the form

$$\widehat{H}^{(0)}(z)=\sum_{m\ge 1}h_m z^{-am},$$

where substitution into (11) gives

$$h_1=-\frac{c_a}{\kappa_a},\qquad h_{m+1}=\frac{P_a(-m)}{\kappa_a}h_m\quad(m\ge 1). \tag{13}$$

This recurrence constructs the unique Laurent-series particular solution. Indeed, the difference of two such solutions would be a Laurent-series solution of the homogeneous equation; its leading monomial could not cancel the exponent-raising term $-\kappa_a z^a H$ in (11).

The homogeneous equation associated with (11) has characteristic equation

$$\lambda^a=\frac{a^a}{(a+1)^{a+1}}.$$

Its roots are the distinct nonzero numbers $\rho_a\zeta$ with $\zeta^a=1$. The corresponding formal solutions have exponential parts $\rho_a\zeta z$ and contain no logarithms because the characteristic numbers are simple; see Salikhov [8, Remark 2, p. 206]. The Levelt–Turrittin theorem therefore decomposes the augmented formal differential module as one rank-one zero-exponential summand and a rank-one summands with exponential parts $\rho_a\zeta z$ [10, Chapter 3, Theorem 3.1]. The zero summand has rank one because the nonzero summands already account for rank a, and the vector

$$\left(1, \widehat{H}^{(0)}, \theta\widehat{H}^{(0)}, \ldots, \theta^{a-1}\widehat{H}^{(0)}\right)^{\mathsf{T}}$$

supplies a nonzero zero-exponential solution.

The direct-sum decomposition gives a K-linear projection onto the zero-exponential summand which commutes with θ. A formal fundamental matrix is generated by Laurent series, formal powers of z, and the exponentials $e^{\rho_a\zeta z}$, without adjoining $\log z$. Salikhov's admissible-field construction treats the logarithm as a separate transcendental generator [8, pp. 207–210]. Hence the formal Picard–Vessiot field generated by this matrix is closed under field operations and differentiation without acquiring powers of $\log z$. The formal image of $\mathcal{K}_a$ lies in this log-free field.

By (7), the zero-exponential component of F_0 is $P_0(\theta)\widehat{H}^{(0)}$ and begins with z^{-a}. Projecting the right side of (10) therefore gives

$$-(a+1)!z^{-a} + O(z^{-2a}).$$

The operator $\theta/a + 1$ annihilates z^{-a}, while

$$\left(\frac{\theta}{a} + 1\right)\left(z^{-a}\log z\right) = \frac{1}{a}z^{-a}.$$

Thus the projected equation forces the term (12). A solution of the homogeneous equation is a constant multiple of z^{-a} and cannot remove the logarithm. Therefore G does not belong to the log-free field $\mathcal{K}_a$. □

The lemma excludes membership in $\mathcal{K}_a$. The following trace argument excludes algebraicity over it.

Lemma 3.2 (Trace descent)**.** *Let K be a differential field of characteristic zero with algebraically closed field of constants, and let $R, p \in K$. Suppose that y is algebraic over K and satisfies $Dy + py = R$. Then the equation has a solution $y_0 \in K$. If every solution of $Du + pu = 0$ in an algebraic extension of K already belongs to K, then $y \in K$.*

Proof. Put $L = K(y)$ and $d = [L : K]$. Derivation commutes with the field trace, so

$$y_0 := \frac{1}{d}\operatorname{Tr}_{L/K}(y)$$

belongs to K and satisfies $Dy_0 + py_0 = R$. The difference $y - y_0$ satisfies the homogeneous equation. A finite algebraic differential extension introduces no new constants because the constant field of K is algebraically closed. The final assertion follows. □

Apply Lemma 3.2 to (10), with $D = \theta/a$ and $p = 1$. The homogeneous solutions are cz^{-a} with $c \in \mathbb{C}$, and already belong to $\mathcal{K}_a$. If G were algebraic over $\mathcal{K}_a$, the trace lemma would imply $G \in \mathcal{K}_a$, contrary to Lemma 3.1. Hence G is transcendental over $\mathcal{K}_a$, and

$$F_0, F_1, \ldots, F_{a-1}, G \tag{14}$$

are algebraically independent over $\mathbb{C}(z)$.

3.3 Specialization and the remaining intercepts

The functions in (14), together with the derivatives needed to close (11) and (10), form a system of E-functions over $\overline{\mathbb{Q}}(z)$ with no nonzero finite singularity. Its function field is $\mathcal{K}_a(G)$ and has transcendence degree $a+1$. Beukers's Theorem 1.1 preserves this transcendence degree at the ordinary algebraic point $z = 1$ [7, Theorem 1.1]. The triangular transformation (7) shows that the corresponding value field is generated by

$$F_0(1), F_1(1), \ldots, F_{a-1}(1), G(1).$$

These values are therefore algebraically independent over $\overline{\mathbb{Q}}$. Equation (3) gives the algebraic independence of

$$C_{a,0}, C_{a,1}, \ldots, C_{a,a-1}, C_{a,a+1}. \tag{15}$$

It remains to express the other intercepts in this basis. Shifting the summation index gives

$$C_{a,a} = (a+1)C_{a,0} - \frac{1}{a!} \tag{16}$$

and, for $b > a+1$,

$$C_{a,b} = \frac{C_{a,b-1} - (a+1)C_{a,b-a-1} + (a+1)/b!}{b-a-1}. \tag{17}$$

Thus every $C_{a,b}$ with $b \geq 0$ is a rational affine combination of (15). For $1 - a \leq b < 0$, set $r = a + 1 + b$, so that $2 \leq r \leq a$. Another index shift, followed by (6), gives

$$C_{a,b} = \frac{1}{r!} + \frac{a+1-r}{a+1}C_{a,r} + \frac{1}{a+1}C_{a,r-1}. \tag{18}$$

Equation (16) eliminates $C_{a,a}$ when $r = a$, proving the affine-span assertion for all admissible intercepts.

These affine combinations are nonconstant. Equation (16) gives an invertible affine transformation from the basis (15) to the block $C_{a,1}, \ldots, C_{a,a+1}$. For each $b > a+1$, equation (17) advances the block $C_{a,b-a-1}, \ldots, C_{a,b-1}$ to the next block. Its linear part is invertible because the coefficient of the discarded first entry in the new final coordinate is $-(a+1)/(b-a-1) \neq 0$. Induction therefore shows that every $C_{a,b}$ with $b \geq 0$ has a nonzero linear part in the basis (15). The expression (18) has a nonzero linear part as well, including when $r = a$ after using (16).

A nonconstant affine combination of algebraically independent numbers is transcendental. Hence every admissible $C_{a,b}$ is transcendental, and the affine span together with (15) gives

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(C_{a,b} : b \geq 1 - a) = a + 1.$$

The same argument shows that the affine recurrences account for all algebraic relations at fixed slope: after any finite collection of the constants is expressed in the basis (15), its ideal of polynomial relations is generated by the resulting affine linear relations. This completes the proof of Theorem 1.

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